

SAM-Wav2lip++: Enhancing Behavioral Realism in Synthetic Agents through Audio-Driven Speech and Action Refinement

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Abstract—Digital human generation is a forward-looking field in technology. Despite significant progress in the generation of speaking facial videos, many challenges remain unaddressed. Issues such as unnatural head movements, distorted expressions, artifacts in generated videos, and uncoordinated limb movements persist. Most current efforts are focused on specific individuals, with enhancements often limited to head movements without further advancing the overall behavioral actions of digital humans. In this context, we introduce a new dataset, CFMD, and a novel model, SAM-Wav2lip++, capable of generating consistent, audio-synchronized lip and behavior action videos from a single reference image of any identity. This work features three main innovative components: (1) a contrastive lip-sync discriminator for precise lip synchronization, (2) a generator for the synthesis of sound-action consistency, and (3) the SAM module for facial refinement operations. Through extensive experiments and user studies, our results demonstrate that our model can synthesize digital human videos of impressively high perceptual quality that accurately sync lip movements and behavioral actions with the input audio, substantially outperforming the state-of-the-art baselines evaluations.

Index Terms—CFMD, SAM-Wav2lip++, Sound-action Sync, Video Generation

I. INTRODUCTION

In recent years, the rapid rise of short videos on platforms such as TikTok and YouTube has markedly increased the demand for quick video content generation. This trend has catalyzed the advancement of novel video editing tools and platforms, and has given birth to a plethora of AI-based automated solutions. These advancements have rendered the creation of content, such as live commerce, unprecedentedly simple and efficient.

The synthesis of digital avatars driven by audio has long been a hot topic, attracting widespread attention from numerous technology companies and research institutions. Audio-driven digital humans can simulate the digitized voices of real people and are generated from audio and a single image. They have extensive applications in fields such as virtual reality, video conferencing, and live commerce. Currently, video generation methods based on Convolutional Neural Networks [1], [2] suffer from unstable training processes and

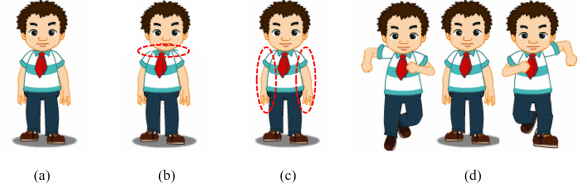


Fig. 1. The figure of (a) represents the reference image used for generating digital human videos; the figure of (b) illustrates a significant issue with current common methods, namely, the possibility of severe head distortion and artifacts during the generation process; the figure of (c) represents the lack of effective constraint and control over body movements in current methods, rendering them incapable of generating body movements; the figure of (d) represents the effects and requirements achievable by our method.

artifacts produced during inference; digital human generation methods based on NeRF [3]–[6] have the disadvantages of high computational resource costs and long training durations; and video generation techniques employing diffusion models [7], [8] are temporarily unfeasible for the creation of digital human videos due to limited duration and weak consistency.

Although the use of deep learning and large-scale datasets has rapidly advanced the field, current datasets predominantly consist of portrait photos. In order to more clearly display the lifelike appearance of virtual digital humans and considering that audio-driven lip synchronization technology is one of the key factors in achieving a highly natural interactive experience, we have collected and shot a vast dataset of model video clips. This dataset is not only massive in size and high in resolution but also consists of full-body shot videos with coordinated speech and motion.

However, in the process of generating audio-driven digital humans, most methods typically only require slight movements of the head [9] or subtle changes in expression, while all other parts remain static; furthermore, there is no good generalization in terms of character identity, as shown in Fig 1. We propose an end-to-end framework called SAM-Wav2lip++, which is based on the original Wav2lip. It enhances the accuracy of the lip-sync discriminator, further improving the consistency of lip movements; replicates the audio-driven action generator, making the body movements in the synthesized videos more coordinated. By synthesizing videos with consistent audio-driven actions, the movements

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of digital humans will be closer to those of real human actions, thereby enhancing the realism and immersion of the videos. We fine-tune the SAM model [10], adding a SAM module for facial refinement, which reduces the generation of head artifacts and distortions compared to the previous SAM. Unlike methods such as Sadtalker and Geneface++, which only involve head movements, our method can generate visually appealing fully animated digital human videos.

The main contributions of this paper can be summarized as follows:

- We have introduced a massive high-definition dataset known as CFMD. This dataset provides us with a wealth of resources, enabling the model to be trained and tested in various high-definition scenarios.
- In order to enhance the realism of the video, we specifically added an audio-driven action generator that can produce behavioral actions perfectly coordinated with the audio, synchronizing the actions in the video with the audio signal.
- In terms of facial refinement processing, we have introduced the SAM module. This module can effectively reduce head artifacts and distortion, significantly enhancing image quality. Simultaneously, the SAM module also enhances the accuracy of lip-sync, making the mouth shape more consistent with the voice, further enhancing the realism and credibility of the video.

II. RELATED WORK

A. Audio-Driven Talking Face Generation

Early work in this field was often constrained by the limited scope for generating identities and the limited lexical choices available. The Barack Obama video-related work [17], [25] made significant progress in generating realistic videos of speaker faces by learning to map input audio to corresponding lip movements. However, these methods are only trained on specific speakers and therefore cannot synthesize new identities or voices. A recent study [26] proposed a method for seamlessly editing videos of a single speaker by adding or removing phrases from the speech. However, this approach requires an hour of data per speaker to accomplish the task. To reduce the data requirements, another study [27] attempted a two-stage approach, where speaker-independent features are first learned, followed by rendering mapping learning using approximately five minutes of data from the target speaker. Nevertheless, this approach employs a smaller corpus size for speaker-independent network training and still requires clean training data for each target speaker, which adds additional overhead. Recent research efforts [27], [28] have been devoted to generating accurate lip movements via perceptual discriminators. Given that real videos contain diverse motions, ATVGnet [29] innovatively employs face landmarks as intermediate representations for generating video frames. MakeItTalk [30] proposes a similar approach, which is unique in that it separates the content and speaker information from the input audio signal. However, due to the highly coupled nature of facial signatures, methods for

generating talking heads in the de-entangled space have gradually gained attention. One such method is PC-AVS [31], which utilizes implicit latent codes to decouple head poses and expressions. However, this approach generates images of low resolution and relies on the control signals from an external video. An external video was utilized by external video. Audio2Head [32] and Wang *et al* [33] were able to successfully generate face images of speaking heads using a video-driven approach [34]. Wav2lip [35] focuses on the precise synchronization of mouth movements with audio signals. Nevertheless, the head movements of these methods are still unprocessed and may result in identity distortion and expression inaccuracies, accompanied by noticeable artifacts. We have introduced the SAM module, a mechanism capable of combining a single reference image of any identity with audio to produce lip movements that are highly synchronized with the audio. This technology ensures an exact match between lip movements and audio, delivering a more natural and authentic visual experience to users.

B. Motion Generation

The current field of digital humans primarily concentrates on the generation of the upper body, while the coordination of whole-body motions, as a novel and under-explored area, is in need of in-depth research. CSGN [36] employs a graph-convolutional GAN framework for generating unconstrained motions, whereas [37] provides an in-depth study of generating dynamically changing motions over an infinite period of time. In addition, numerous studies have explored motion generation under different conditions. These include music-driven dance motion synthesis [38]–[40] and motion generation based on linguistic descriptions [41]–[44]. Recently, proposed text-driven diffusion models [45], [46] for human motion generation. In the context of action-conditional generation, Action2Motion [47] employs a GRU-time VAE to achieve frame-by-frame motion generation, whereas ACTOR [48] integrates VAE with Transformer to learn sequence-level latent distributions. ActFormer [49] also accepts sequence-level latent priors as inputs, but in contrast to ACTOR, its inputs originate from GP-prior sampled sequence of latent vectors, which enhances temporal correlation and simplifies the real motion generation process. To further enhance the coordination of full-body movements in digital humans and match these with face generation, we utilized GAN-based 3D convolution techniques to generate behavioral actions. This innovative approach has significantly advanced research in the field.

III. CHINESE FASHION MOTE DATASET

A. CFMD Dataset

This dataset is a high-quality, research-oriented resource that amasses data from over 8000 speakers with a storage size of 872GB. It comprehensively encompasses the Chinese HD Talking Face Database, which is hosted by the Image Institute of Shanghai Jiao Tong University. To ensure the richness and diversity of the data, speakers from different age groups, genders, dialect regions, as well as a variety of voice

TABLE I
COMPARISON OF THE CFMD DATASET WITH OTHER DATASETS.

Name	Year	Total Hours	Speaker	Movement	Full-body Shot	Source	Language	Resolution
GRID [11]	2006	27.5	33	No	No	Lab	English	720 × 576
CREMA-D [12]	2014	11.1	91	Yes	No	Lab	English	1280 × 720
MSP-IMPROV [13]	2016	18	12	Yes	No	Lab	English	-
LRW [14]	2017	173	1k+	No	No	BBC	English	-
LRS2-BBC [15]	2018	224.5	1k+	Yes	No	BBC	English	-
LRS3-TED [16]	2018	438	5k+	Yes	No	TED	English	224 × 224
ObamaSet [17]	2017	14	1	Yes	No	Internet	English	-
VoxCeleb1 [18]	2017	352	1.2k	Yes	No	YouTube	English	-
VoxCeleb2 [19]	2018	2.4k	6.1k+	Yes	No	YouTube	English	-
AVSpeech [20]	2018	4.9k	150k+	Yes	No	YouTube	Multiple	-
LRW-1000 [21]	2019	57	2k+	Yes	No	Internet	Chinese	-
FaceForensics++ [22]	2019	5.7	1k	Yes	No	Internet	English	-
MEAD [23]	2020	39	60	Yes	No	Lab	English	1920 × 1080
HDTF [24]	2021	15.8	362	Yes	No	Internet	English	512 × 512
CFMD(Ours)	2024	382	8k+	Yes	Yes	Lab	Chinese	2k/4k



Fig. 2. Some video data set sample images.

intonations and facial expressions are included. Thereby, it provides a wealth of testing and training data for various applications. The video data set is shown in Fig 2, and the face area has been mosaiced. We also compare our dataset with other datasets, and the results show that the CFMD dataset ranks at the forefront in several indicators, such as the number of people and duration, as shown in Table I. Since this data set involves the privacy of the speaker's portrait, it can only be made public after communicating with the speaker.

B. Data Collection

Constructing a precise and high-quality dataset is essential for model learning and training. This dataset, by inviting professional models for on-site shoots, ensures the diversity and expressiveness of individuals in the dataset. Through meticulous preparation and rigorous processes, it captures

high-quality videos that reflect people's diversity and dynamic changes.

Model Invitation and Selection. To ensure the diversity of the dataset, the selection of models fully utilizes diversity and inclusiveness, covering different ages, skin colors, body types, and styles. Professionalism, portfolio quality, and adaptability become the criteria for this dataset, ensuring its comprehensiveness and applicability.

Scene Setting. The shooting scene is established in a professional photography studio, using green screen technology to simplify the background replacement in the post-processing phase, eliminating the need to actually change the site setting. This offers great flexibility and reduces costs.

Lighting Setting. In any high-quality video production, lighting is a crucial element. Experienced lighting technicians are chosen for the shooting process to arrange the lighting, ensuring that the contours of the face and body are appropriately highlighted while avoiding shadows and glare that could cause segmentation errors. High-quality lighting settings are not only vital for the video effect but also directly impact the algorithm's segmentation accuracy.

Photography Equipment. The selection of photography equipment considers the importance of high resolution and high frame rate for the video dataset. Therefore, high-end photography equipment, including 4K or even 8K cameras, is used during the shooting process to capture the finest details. Moreover, lens choices and camera settings need to be adjusted by experienced photographers to adapt to diverse shooting conditions.

Shooting Process. The shooting process is the most critical part of this data collection. It requires models to perform various poses and expressions, following the director's guidance to complete a series of predetermined actions. This ensures that each action is clearly captured and has sufficient variation. Continuous repetition and adjustment are the norms during the shooting process to ensure the acquisition of images that meet the requirements.

IV. METHOD

The core architecture of the model is summarized as the "Lip-Sync Action Generation Discrimination Mechanism

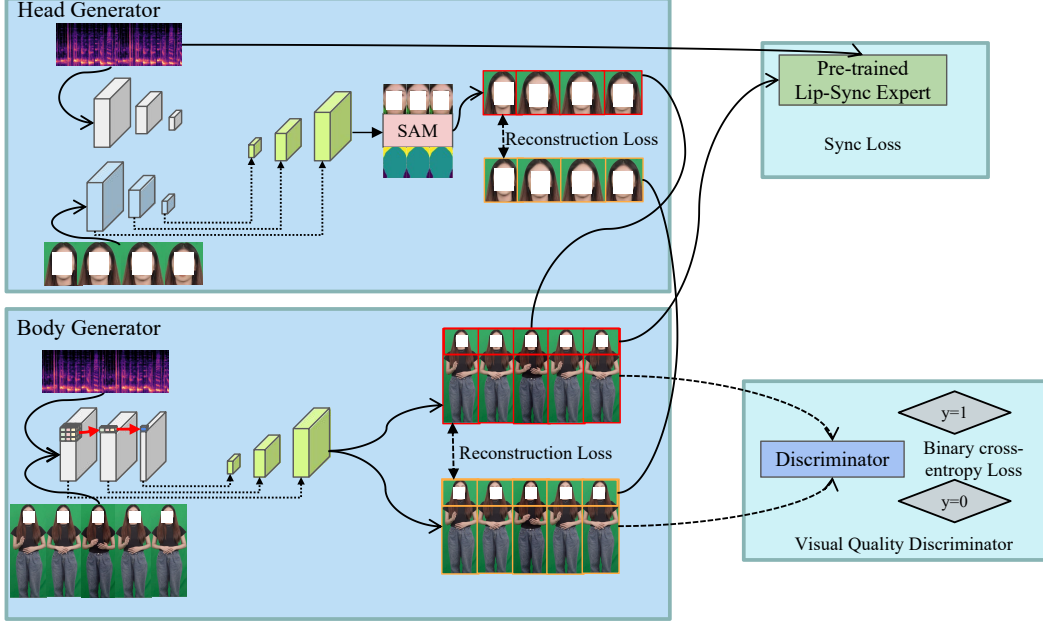


Fig. 3. The overall architecture of the model. The generator primarily divides into a head generator and a body generator. Sync Loss from the Lip-Sync Expert discriminator, combined with the facial reconstruction loss through the SAM module and the binary loss from the Visual Quality Discriminator, serves as the model’s loss function. The Visual Quality Discriminator enhances the visual quality and synchronicity by treating the head face and limb behaviors as a whole.

Model”. The generator is mainly divided into a head generator and a body generator. The head generator section is enhanced with the SAM module to improve facial replication accuracy, while the body generator is responsible for generating limb behavior actions. The Lip-Sync Expert is a pre-trained model used to detect lip-sync errors generated by the generator. The Visual Quality Discriminator enhances visual quality and synchronization accuracy by treating the head face and limb behaviors as a whole. The overall architecture of the model is shown in Fig 3. Human faces are blurred to conceal identity. The same applies hereinafter.

A. Lip-Sync

In our experiments, although the lip-sync discriminator could generate accurate lip shapes, it sometimes led to slightly blurred deformation areas or minor artifacts. To mitigate this slight loss in quality, we incorporated the SAM module. In this section, we used CelebAHairMask-HQ to fine-tune SAM, and the fine-tuning results showed in in Fig 4. The visual quality discriminator provides constraints on the refined faces and generation, especially penalizing unrealistic faces. The calculation method for the synchronization probability of input audio-video pairs is by calculating the dot product between the entire video and the audio embeddings v and s . It generates a single value ranging within $[0, 1]$ for each sample, which serves to represent the probability of synchronization and can accurately reflect the

degree of synchronization between audio and video:

$$P_{sync} = \frac{v \cdot s}{\max(\|v\|_2 \cdot \|s\|_2, \epsilon)} \quad (1)$$

where P_{sync} represents the probability of synchronization of the input audio-video pair.

The SAM module is responsible for refining the generated video frames, while the trained generator performs optimization operations with the goal of minimizing the L1 loss between the refined frame L_{SAM} and the ground truth L_{GThead} .

$$L_{ReconHead} = \frac{1}{N} \sum_{i=1}^N \|L_{SAM} - L_{GThead}\|_1 \quad (2)$$

The head generator model also aims to minimize the expert sync loss as an optimization objective:

$$L_{sync} = \frac{1}{N} \sum_{i=1}^N -\log(P_{sync}^i) \quad (3)$$

where P_{sync} is calculated from Equation 1. N represents the batchsize, and i represents a certain frame of the video clip.

B. Action Generation

In relation to Wav2lip, we have decoupled the motion posture module from the overall architecture, with a particular focus on capturing dynamic human information. We have abandoned previous 2D convolution in favor of a



Fig. 4. The results of fine-tuning SAM using CelebAHairMask-HQ.

3D convolution form that combines 2D and 1D. This key part involves extracting temporally continuous and spatially detailed features to capture the natural movements of the subject. We considered not only the posture at each moment but, more importantly, analyzed the temporal changes in posture to generate motion sequences that appear more natural and fluid. Based on this, we have adopted an advanced generative adversarial network architecture, particularly introducing the Self-Attention Mechanism. The Self-Attention Mechanism significantly enhances the model’s ability to capture long-distance interactions by modeling global dependencies, which is crucial for managing temporal relationships within video sequences. Specifically, within the generative network, the self-attention layer assists the model in considering information from other frames when generating a certain frame, thereby producing temporally continuous and spatially detailed video sequences. Additionally, the model computes reconstruction loss for the entire image to ensure the generated posture is correct, the identity is preserved and even the surrounding background of the face.

Body Generator targets the L1 loss between the generated video frame L_{gen} and the real situation L_{gt} , and performs optimization operations:

$$L_{ReconBody} = \frac{1}{N} \sum_{i=1}^N \|L_{Gen} - L_{GTBody}\|_1 \quad (4)$$

Although actions and faces are generated independently, in order to ensure the consistency and coordination of the generation quality, we calculate and organize the weighted sum of losses.

$$L_{total} = (1 - s_w - s_g) \cdot (L_{ReconHead} + L_{ReconBody}) + s_w \cdot L_{sync} + s_g \cdot L_{ReconHead} + L_{ReconBody} \quad (5)$$

where s_w is the synchronization penalty weight, and s_g is the adversarial loss, we still design it according to the parameters of wav2lip.

V. EXPERIMENT AND EVALUATION

The generation of digital human videos is a complex and challenging endeavor, with the overarching objective of producing realistic and compelling image sequences that depict humans in motion, indistinguishable from actual video footage. However, assessing the quality and authenticity of the generated video content is not a straightforward task, owing to the high degree of subjectivity in human perception



Fig. 5. (a1) exhibits noticeable artifacts, which detract from the overall visual experience. (a2) shows significant distortion, reducing the overall quality of the video generation. The motion in (b1) is quite monotonous and has poor synchronization with the audio, diminishing the realism of the video. (b2) hardly displays any meaningful motion, failing to effectively convey the audio content. (a3) and (b3) demonstrate the effectiveness of our model, not only addressing the issues of artifacts and distortion but also generating actions that are highly synchronized and naturally fluid with the audio.

and the technical diversity of the metrics used to measure video quality.

To conduct a comprehensive evaluation of the generated videos, it is essential to assess them from multiple perspectives. Therefore, we have bifurcated the evaluation stage into a quantitative assessment of video quality and a qualitative, human-centric subjective assessment.

TABLE II
RESULTS ON QUANTITATIVE EVALUATION SCORES.

Models	PSNR $\uparrow$	SSIM $\uparrow$	LMD $\downarrow$	FVD $\downarrow$
Speech2Vid	14.87	0.6954	0.57	20.57
LipGAN	27.45	0.8624	0.54	14.68
Wav2lip	<u>35.42</u>	0.9751	<u>0.23</u>	11.84
SadTalker	34.65	<u>0.9822</u>	0.46	<u>10.47</u>
SAM-Wav2lip++	37.96	0.9865	0.14	10.28

A. Experimental Results

We mainly compared three models: Wav2Lip, SadTalker, and SAM-Wav2Lip++, as shown in Fig 5. It can be observed that the Wav2Lip model exhibits obvious artifact issues on the CFMD dataset. These artifacts not only affect the accuracy of lip-sync but also reduce the quality of the overall video generation. On the other hand, the SadTalker model displays significant distortion at the joints, making the generated movements look unnatural, and it also fails to effectively generate actions that match the speech.

In contrast, our SAM-Wav2Lip++ model has achieved significant improvements in several aspects. It successfully addresses the artifact issues of Wav2Lip, making the generated lip movements clearer, more natural, and highly synchronized with the audio signal. At the same time, SAM-Wav2Lip++ overcomes the distortion problems at the joints seen with SadTalker, and by introducing advanced algorithms and technologies, it makes the generated behavioral actions smoother and more realistic.

TABLE III
RESULTS ON HUMAN EVALUATION SCORES.

Models	Number of videos generated	Sync Acc $\uparrow$	Visual Qual $\uparrow$	Overall Experience $\uparrow$	Preference $\uparrow$
Speech2Vid	200	1.41	1.23	0.92	-0.54
LipGAN	200	2.98	3.91	3.45	-0.34
Wav2lip	200	4.13	3.87	4.23	0.24
SadTalker	200	4.26	3.62	4.22	0.43
SAM-Wav2lip++	200	4.32	4.02	4.35	0.76

B. Quantitative Analysis

In quantitative evaluation, our focus is on objective metrics that can be consistently measured across different videos. This includes standard video quality metrics such as PSNR, SSIM, LMD, FID, and FVD. These metrics respectively reflect the quality characteristics of the video from different dimensions, helping us to comprehensively and accurately assess the generation effect of digital human videos. PSNR measures pixel-level differences, SSIM focuses on structural similarity, LMD evaluates the coherence of motion information, FID measures the distribution differences in the feature space, and FVD pays attention to the continuity of video dynamic information. By using these metrics together, we are able to obtain a comprehensive evaluation of the quality of digital human videos. The quantitative evaluation results are shown in Table II.

- Peak Signal-to-Noise Ratio (PSNR) is a fundamental metric for measuring the quality of images or videos, based on the calculation of errors between pixels. In the assessment of audio-driven digital human videos, PSNR aids in understanding the degree of distortion at the pixel level. Higher PSNR values typically indicate better video quality, with complete preservation of details and minimal distortion.

$$PSNR = 10 \cdot \log_{10} \left(\frac{MAX^2}{MSE} \right) \quad (6)$$

where MAX represents the maximum possible pixel value and MSE is the mean square error.

- Structural Similarity Index Measure (SSIM) is a quality assessment metric based on the similarity of image structures, taking into account luminance, contrast, and structural information. In the evaluation of digital human videos, SSIM is capable of capturing the sensitivity of the human eye to structural information, thus reflecting the quality of the video more accurately.

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (7)$$

where x and y are the two images to be compared, μ_x and μ_y are the mean brightness values of images x and y respectively, σ_x^2 and σ_y^2 are the mean brightness values of images x and y respectively, σ_{xy} is the brightness covariance between images x and y , C_1 and C_2 are constants used to stabilize calculations.

- Local Motion Difference (LMD) is an index used to evaluate the continuity of motion information in videos.

LMD assesses the performance of generated videos in terms of motion information by calculating local motion differences between adjacent frames within the video. Low LMD values imply a more coherent motion information in the generated video, contributing to the enhancement of overall visual perception.

$$LMD = \frac{1}{N} \sum_{i=1}^N \sqrt{(x_{g_i} - x_{r_i})^2 + (y_{g_i} - y_{r_i})^2} \quad (8)$$

where x_{g_i} and y_{g_i} are respectively the x and y coordinates of the i-th feature point on the generated image, x_{r_i} and y_{r_i} are respectively the x and y coordinates of the i-th feature point on the real image.

- Frechet Video Distance (FVD) is a metric specifically used to evaluate the quality of video sequences. FVD can measure the coherence and authenticity of dynamic information in generated videos. By calculating the distance between generated and real videos in the feature space, FVD can reflect the overall dynamic performance of the generated videos.

$$FVD = \|m_1 - m_2\|^2 + Tr(C_1 + C_2 - 2(C_1 C_2)^{1/2}) \quad (9)$$

where $\|m_1 - m_2\|^2$ is the square of the Euclidean distance between two mean vectors, Tr represents the trace of a matrix, $C_1 + C_2 - 2(C_1 C_2)^{1/2}$ calculates the "distance" of the two covariance matrices.

C. Human evaluation

Although quantitative analysis metrics are useful, they also have limitations as they cannot fully capture the human perception of realism in videos. Advanced digital human video generation systems can score high on these metrics while still producing videos that look unnatural or contain visual anomalies detectable by the human eye.

To address these limitations, we have also incorporated qualitative assessments into our evaluation protocol. This involves human subjects assessing various aspects of video realism, such as the plausibility of character movements, consistency of characters, and fidelity. Subjective evaluation emphasizes the viewer's perception, which is the ultimate test of the realism of digital humans in video content. We conducted 1000 instances of digital human video generation based on reference images and audio, and selected 30 students from different laboratories to score them. The scoring criteria included Sync Accuracy (Sync Acc), Visual Quality (Visual Qual), Overall Experience, and Preference. The scoring range for the first three criteria is [0,5], while

the range for the last criterion is $[-1,1]$, representing a scale from dislike to like, as demonstrated in Table III. It can be seen that our model surpasses the previous models in every evaluated criterion.

VI. CONCLUSION

In this study, we meticulously crafted a comprehensive and refined digital human video generation dataset, which not only encompasses subtle facial expressions and the speaker's vocal information but also integrates a wealth of bodily movements, all presented in high-definition quality. Compared to existing datasets, ours demonstrates significant advantages across a range of key metrics. Building on this, we innovatively propose a model that first drives the generation of the head and facial components using audio, then leverages the advanced SAM framework to significantly improve the precision of the generated content, forming a perfect complementarity with the Lip-Sync discriminator. In the audio-driven behavior motion generation phase, we cleverly employed 3D temporal convolutional techniques to ensure that the produced actions are both accurate and natural. Through rigorous quantitative analysis and human assessment, our model has surpassed existing methods on multiple key evaluation metrics. We have reason to believe that the dataset and model we provide will set a new milestone in the field of digital human generation and inject powerful momentum into the development of this field.

REFERENCES

- [1] A. Jamaludin, J. S. Chung, and A. Zisserman, "You said that?: Synthesizing talking faces from audio," *International Journal of Computer Vision*, vol. 127, pp. 1767–1779, 2019.
- [2] Z. Ye, Z. Zhao, Y. Ren, and F. Wu, "Syntaspeech: Syntax-aware generative adversarial text-to-speech," *arXiv preprint arXiv:2204.11792*, 2022.
- [3] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, "Nerf: Representing scenes as neural radiance fields for view synthesis," *Communications of the ACM*, vol. 65, no. 1, pp. 99–106, 2021.
- [4] Y. Guo, K. Chen, S. Liang, Y.-J. Liu, H. Bao, and J. Zhang, "Ad-nerf: Audio driven neural radiance fields for talking head synthesis," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 5784–5794.
- [5] Z. Ye, Z. Jiang, Y. Ren, J. Liu, J. He, and Z. Zhao, "Geneface: Generalized and high-fidelity audio-driven 3d talking face synthesis," *arXiv preprint arXiv:2301.13430*, 2023.
- [6] Z. Ye, J. He, Z. Jiang, R. Huang, J. Huang, J. Liu, Y. Ren, X. Yin, Z. Ma, and Z. Zhao, "Geneface++: Generalized and stable real-time audio-driven 3d talking face generation," *arXiv preprint arXiv:2305.00787*, 2023.
- [7] A. Blattmann, T. Dockhorn, S. Kulal, D. Mendelevitch, M. Kilian, D. Lorenz, Y. Levi, Z. English, V. Voleti, A. Letts *et al.*, "Stable video diffusion: Scaling latent video diffusion models to large datasets," *arXiv preprint arXiv:2311.15127*, 2023.
- [8] S. Zhang, J. Wang, Y. Zhang, K. Zhao, H. Yuan, Z. Qin, X. Wang, D. Zhao, and J. Zhou, "I2vgen-xl: High-quality image-to-video synthesis via cascaded diffusion models," *arXiv preprint arXiv:2311.04145*, 2023.
- [9] W. Zhang, X. Cun, X. Wang, Y. Zhang, X. Shen, Y. Guo, Y. Shan, and F. Wang, "Sadtalker: Learning realistic 3d motion coefficients for stylized audio-driven single image talking face animation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 8652–8661.
- [10] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo *et al.*, "Segment anything," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 4015–4026.
- [11] M. Cooke, J. Barker, S. Cunningham, and X. Shao, "The grid audio-visual speech corpus," 2006.
- [12] H. Cao, D. G. Cooper, M. K. Keutmann, R. C. Gur, A. Nenkova, and R. Verma, "Crema-d: Crowd-sourced emotional multimodal actors dataset," *IEEE transactions on affective computing*, vol. 5, no. 4, pp. 377–390, 2014.
- [13] C. Busso, S. Parthasarathy, A. Burmanian, M. AbdelWahab, N. Sadoughi, and E. M. Provost, "Msp-improv: An acted corpus of dyadic interactions to study emotion perception," *IEEE Transactions on Affective Computing*, vol. 8, no. 1, pp. 67–80, 2016.
- [14] J. S. Chung and A. Zisserman, "Lip reading in the wild," in *Computer Vision—ACCV 2016: 13th Asian Conference on Computer Vision, Taipei, Taiwan, November 20–24, 2016, Revised Selected Papers, Part II* 13. Springer, 2017, pp. 87–103.
- [15] T. Afouras, J. S. Chung, A. Senior, O. Vinyals, and A. Zisserman, "Deep audio-visual speech recognition," *IEEE transactions on pattern analysis and machine intelligence*, vol. 44, no. 12, pp. 8717–8727, 2018.
- [16] T. Afouras, J. S. Chung, and A. Zisserman, "Lrs3-ted: a large-scale dataset for visual speech recognition," *arXiv preprint arXiv:1809.00496*, 2018.
- [17] S. Suwajanakorn, S. M. Seitz, and I. Kemelmacher-Shlizerman, "Synthesizing obama: learning lip sync from audio," *ACM Transactions on Graphics (TOG)*, vol. 36, no. 4, pp. 1–13, 2017.
- [18] A. Nagrani, J. S. Chung, and A. Zisserman, "Voxceleb: a large-scale speaker identification dataset," *arXiv preprint arXiv:1706.08612*, 2017.
- [19] J. S. Chung, A. Nagrani, and A. Zisserman, "Voxceleb2: Deep speaker recognition," *arXiv preprint arXiv:1806.05622*, 2018.
- [20] A. Ephrat, I. Mosseri, O. Lang, T. Dekel, K. Wilson, A. Hassidim, W. T. Freeman, and M. Rubinstein, "Looking to listen at the cocktail party: A speaker-independent audio-visual model for speech separation," *arXiv preprint arXiv:1804.03619*, 2018.
- [21] S. Yang, Y. Zhang, D. Feng, M. Yang, C. Wang, J. Xiao, K. Long, S. Shan, and X. Chen, "Lrw-1000: A naturally-distributed large-scale benchmark for lip reading in the wild," in *2019 14th IEEE international conference on automatic face & gesture recognition (FG 2019)*. IEEE, 2019, pp. 1–8.
- [22] A. Rossler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, "Faceforensics++: Learning to detect manipulated facial images," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 1–11.
- [23] K. Wang, Q. Wu, L. Song, Z. Yang, W. Wu, C. Qian, R. He, Y. Qiao, and C. C. Loy, "Mead: A large-scale audio-visual dataset for emotional talking-face generation," in *European Conference on Computer Vision*. Springer, 2020, pp. 700–717.
- [24] Z. Zhang, L. Li, Y. Ding, and C. Fan, "Flow-guided one-shot talking face generation with a high-resolution audio-visual dataset," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 3661–3670.
- [25] R. Kumar, J. Sotelo, K. Kumar, A. De Brebisson, and Y. Bengio, "Obamanet: Photo-realistic lip-sync from text," *arXiv preprint arXiv:1801.01442*, 2017.
- [26] J. Thies, M. Elgharib, A. Tewari, C. Theobalt, and M. Nießner, "Neural voice puppetry: Audio-driven facial reenactment," in *Computer Vision—ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XVI* 16. Springer, 2020, pp. 716–731.
- [27] K. Cheng, X. Cun, Y. Zhang, M. Xia, F. Yin, M. Zhu, X. Wang, J. Wang, and N. Wang, "Videoretalking: Audio-based lip synchronization for talking head video editing in the wild," in *SIGGRAPH Asia 2022 Conference Papers*, 2022, pp. 1–9.
- [28] P. KR, R. Mukhopadhyay, J. Philip, A. Jha, V. Nambodiri, and C. Jawahar, "Towards automatic face-to-face translation," in *Proceedings of the 27th ACM international conference on multimedia*, 2019, pp. 1428–1436.
- [29] L. Chen, R. K. Maddox, Z. Duan, and C. Xu, "Hierarchical cross-modal talking face generation with dynamic pixel-wise loss," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 7832–7841.
- [30] Y. Zhou, X. Han, E. Shechtman, J. Echevarria, E. Kalogerakis, and D. Li, "Makeltalk: speaker-aware talking-head animation," *ACM Transactions On Graphics (TOG)*, vol. 39, no. 6, pp. 1–15, 2020.
- [31] H. Zhou, Y. Sun, W. Wu, C. C. Loy, X. Wang, and Z. Liu, "Pose-controllable talking face generation by implicitly modularized audio-visual representation," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 4176–4186.

- [32] S. Wang, L. Li, Y. Ding, C. Fan, and X. Yu, "Audio2head: Audio-driven one-shot talking-head generation with natural head motion," *arXiv preprint arXiv:2107.09293*, 2021.
- [33] S. Wang, L. Li, Y. Ding, and X. Yu, "One-shot talking face generation from single-speaker audio-visual correlation learning," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 3, 2022, pp. 2531–2539.
- [34] A. Siarohin, S. Lathuilière, S. Tulyakov, E. Ricci, and N. Sebe, "First order motion model for image animation," *Advances in neural information processing systems*, vol. 32, 2019.
- [35] K. Prajwal, R. Mukhopadhyay, V. P. Nambodiri, and C. Jawahar, "A lip sync expert is all you need for speech to lip generation in the wild," in *Proceedings of the 28th ACM international conference on multimedia*, 2020, pp. 484–492.
- [36] S. Yan, Z. Li, Y. Xiong, H. Yan, and D. Lin, "Convolutional sequence generation for skeleton-based action synthesis," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 4394–4402.
- [37] Y. Zhang, M. J. Black, and S. Tang, "Perpetual motion: Generating unbounded human motion," *arXiv preprint arXiv:2007.13886*, 2020.
- [38] H.-Y. Lee, X. Yang, M.-Y. Liu, T.-C. Wang, Y.-D. Lu, M.-H. Yang, and J. Kautz, "Dancing to music," *Advances in neural information processing systems*, vol. 32, 2019.
- [39] J. Li, Y. Yin, H. Chu, Y. Zhou, T. Wang, S. Fidler, and H. Li, "Learning to generate diverse dance motions with transformer," *arXiv preprint arXiv:2008.08171*, 2020.
- [40] R. Huang, H. Hu, W. Wu, K. Sawada, M. Zhang, and D. Jiang, "Dance revolution: Long-term dance generation with music via curriculum learning," *arXiv preprint arXiv:2006.06119*, 2020.
- [41] A. S. Lin, L. Wu, R. Corona, K. Tai, Q. Huang, and R. J. Mooney, "Generating animated videos of human activities from natural language descriptions," *Learning*, vol. 1, no. 2018, p. 1, 2018.
- [42] A. Ghosh, N. Cheema, C. Oguz, C. Theobalt, and P. Slusallek, "Synthesis of compositional animations from textual descriptions," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 1396–1406.
- [43] H. Ahn, T. Ha, Y. Choi, H. Yoo, and S. Oh, "Text2action: Generative adversarial synthesis from language to action," in *2018 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2018, pp. 5915–5920.
- [44] C. Ahuja and L.-P. Morency, "Language2pose: Natural language grounded pose forecasting," in *2019 International Conference on 3D Vision (3DV)*. IEEE, 2019, pp. 719–728.
- [45] M. Zhang, Z. Cai, L. Pan, F. Hong, X. Guo, L. Yang, and Z. Liu, "Motiondiffuse: Text-driven human motion generation with diffusion model," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2024.
- [46] G. Tevet, S. Raab, B. Gordon, Y. Shafir, D. Cohen-Or, and A. H. Bermano, "Human motion diffusion model," *arXiv preprint arXiv:2209.14916*, 2022.
- [47] C. Guo, X. Zuo, S. Wang, S. Zou, Q. Sun, A. Deng, M. Gong, and L. Cheng, "Action2motion: Conditioned generation of 3d human motions," in *Proceedings of the 28th ACM International Conference on Multimedia*, 2020, pp. 2021–2029.
- [48] M. Petrovich, M. J. Black, and G. Varol, "Action-conditioned 3d human motion synthesis with transformer vae," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 10 985–10 995.
- [49] L. Xu, Z. Song, D. Wang, J. Su, Z. Fang, C. Ding, W. Gan, Y. Yan, X. Jin, X. Yang *et al.*, "Actformer: A gan-based transformer towards general action-conditioned 3d human motion generation," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 2228–2238.